

TABLE 3
MODES OF OPERATION

TIME IN HOURS				6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	00	1	2	3	4	5
MSB		LSB																									
S3	S2	S1																									
PIN 2	PIN 3	PIN 4	MODE																								
0	0	0	1	NIGHT MODE																							
0	0	1	2	NIGHT SMART TOGGLE																							
0	1	0	3	NIGHT SMART WITH BUZZER																							BUZZER AT 05:59:55 sec
0	1	1	4	OFFICE MODE																							
1	0	0	5	OFFICE TOGGLE																							
1	0	1	6	HALL MODE																							
1	1	0	7	ALL DAY TOGGLE MODE																							

allows for monitoring of time and working mode.

The following functions and libraries are used.

Serial.begin(). Establishes serial communication between Arduino Uno board and another device, via a USB cable. It permits the two devices to communicate using a serial protocol. Here serial communication is used only to just read the data from Arduino Uno onto the screen. The displayed data allows us to know the time and working mode.

The following header files are used in the main code.

#include <Wire.h>. The Wire library allows to communicate through I2C devices, also called '2 wire' or 'TWI' (two wire interface). The SDA (data line) and SCL (clock line) are used for communicating with DS1307 RTC module.

#include "RTClib.h". Used for communicating with the DS1307 RTC module. In the next step, initialise the relay pin as output.

Serial.begin(9600). Used for serial communication. For displaying the time on the serial screen, it is

used for just reference purpose.

Two methods of setting the time into the RTC module are:

Method 1.

`rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));` This line sets the time according to your computer time. It will set the present clock time of your system into the DS1307 I2C RTC module.

Method 2.

`rtc.adjust(DateTime(2021, 7, 21, 10, 0, 0));` This command will set the time into the RTC module in this format (year, month, date, hours, minutes, seconds.) (for example: 2021

, July ,21 , 10:00:00 am). Once the time is set, the backup battery in the RTC module keeps the time for a long time.

Construction and testing

Upload the source code into the Arduino Uno board, ensuring proper timing settings as indicated in the source code. Assemble the circuit on a breadboard and connect the 12V DC power adaptor to the barrel jack on the Arduino. Select modes using switches S1 through S3. The relay will energise and de-energise based on the selected modes, controlling the liquid mosquito repellent accordingly.

For manual operation or bypassing the circuit, remove the 12V power supply from the Arduino Uno board and use switch S4. Close switch S4 for manual activation and open it for manual deactivation of the liquid mosquito repellent. **EFY**

EFY Note

The source code of this project is available for free download at www.electronicshobby.com



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